

5.4

Slicing and Dicing using SAP Analysis for Excel

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OBJECTIVE

- To derive business intelligence from a data warehouse – SAP BW/4HANA Composite Provider.
- Create queries in Eclipse and then analyze the data in SAP Analysis for Excel.

ACTIVITIES

- Connect to a composite provider
- Use Characteristics and Measures
- Create crosstabs.
- Use OLAP techniques. Use pivoting, sorting, filtering, hierarchies
- Create conditional formatting
- Create charts and presentations
- Discover data trends

SOFTWARE PREREQUISITES

- Microsoft Windows 10 or higher
- Eclipse with SAP BW Modeling tools installed
- SAP Analysis 2.8 (or higher) for Microsoft Excel
- Microsoft Excel 2016 or higher
- Microsoft PowerPoint 2016 or higher
- SAP GUI 7.70 for Windows

UCC PRODUCTS REQUIRED

- Access to an SAP BW/4HANA 2.0 system
- Access to a *GB Sales Data Composite Provider* within SAP BW/4HANA 2.0

Scenario

As a business analyst for Global Bike Company (GB), you have been given the task of exploring the GB data warehouse by creating queries that discover critical business information that can be used for decision making. Data warehouses provide you with a **single source of truth** by extracting, transforming and loading all relevant business data into one comprehensive InfoProvider. Historical sales data for GB has been modeled and loaded into a composite provider. This will be the data source for your query. Using SAP Analysis, you will use powerful OLAP techniques to perform data analysis and visualization.

Designing Queries in BW Query Designer (Eclipse)

The *GB Sales Data composite provider* is an InfoProvider for several analytics activities that use SAP BusinessObjects tools. The sales data within this composite provider has some interesting and potentially informative trends and it is our goal to discover these trends. Trend analysis can be used to make informed business decisions and plan our strategy for marketing, product development, customer service etc. *Discovery of hidden trends, associations, behavior, and patterns is the main objective of business intelligence and data mining activities.*

A query is often built to access data in the composite provider. For our exercise, Eclipse will be used to build this query. Once saved, the query definition will be available in the SAP BW system for future use.

1. To open Eclipse go to **Start → Programs → Eclipse**
2. Double click on the Project for BW (instructions provided by your instructor if a project is not already listed in Eclipse)
3. Login (contact your UCC for accounts)
4. Go to **BW repository → University Alliances BW/4HANA 2.0 Curriculum**
5. Go to **Your class's InfoArea → find the premade Composite Provider (XXXCP1000 Sales Data Composite Provider)**. This is the composite provider for Global Bike Sales Data from 2007 – 2019 for your class.
6. Right-click the composite provider, then **New Query**.
7. Name: **XXXCP1MQ1###**, Description: **XXX GB Sales Query for Analysis ###**
8. Finish
9. Explore the various panels shown in **Error! Reference source not found..** *Note the various tabs. These tabs will be used to define and customize the query.*

General: MU0CP1MQ1000 - MU0 GB Sales Query for Analysis 000 Active Version

General

Technical Name: MU0CP1MQ1000

Description:

[InfoProvider:](#) MU0CP1

Key Date: Edit...

Output Settings

☒ Adjust Formatting after Refreshing

☐ Suppress Repeated Key Values

☐ Show Scaling for Measures

Sign Format:

Zero Format:

Custom Zero Format:

Result Location

Rows: ☐ Above ☒ Below Columns: ☐ Left ☒ Right

Zero Suppression

Applies to: ☐ Rows ☐ Columns

Condition:

Remote Access

☐ By OLE DB for OLAP

☐ By OData

☐ External SAP HANA View

Universal Display Hierarchy

☐ Active on rows
Expand to Level:

☐ Active on columns
Expand to Level:

[Variables Order](#) [Planning](#)

[Extended](#)

General | **Filter** | Sheet Definition | Conditions | Exceptions | Dependencies | Runtime Properties

Figure 1: Eclipse Query Designer

10. In the left panel, click on **InfoProvider** (Figure).

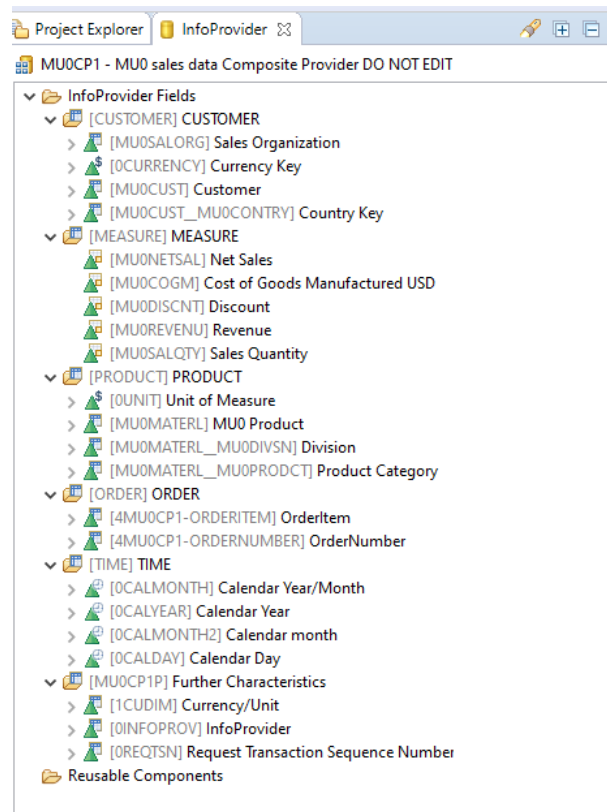


Figure 2: The InfoProvider panel

11. You will see the **key figures** and **characteristics** of the GB composite provider.

Designing your Query

1. Open the **Sheet Definition** tab. You will see **Rows** and **Columns** panels.
2. From the **Product** group, drag the **Product** characteristic and drop it into the **Rows**. Move **Sales Organization** and **Customer** from **Customer** to **Rows**.
3. From the **Key Figures** folder, transfer all of the key figures (**CoGM**, **Discount**, **Net Sales**, **Revenue** and **Sales Quantity**) into **columns**.
4. Free characteristics are characteristics that are *available* to be included in your query results in Analysis but are not displayed in the crosstab when you first run the query. Move the following characteristics into **Free** characteristics – **Division**, **Product Category**, **Country**, **Calendar Year/Month**, **Calendar Year**, and **Calendar Month** (Figure 3).

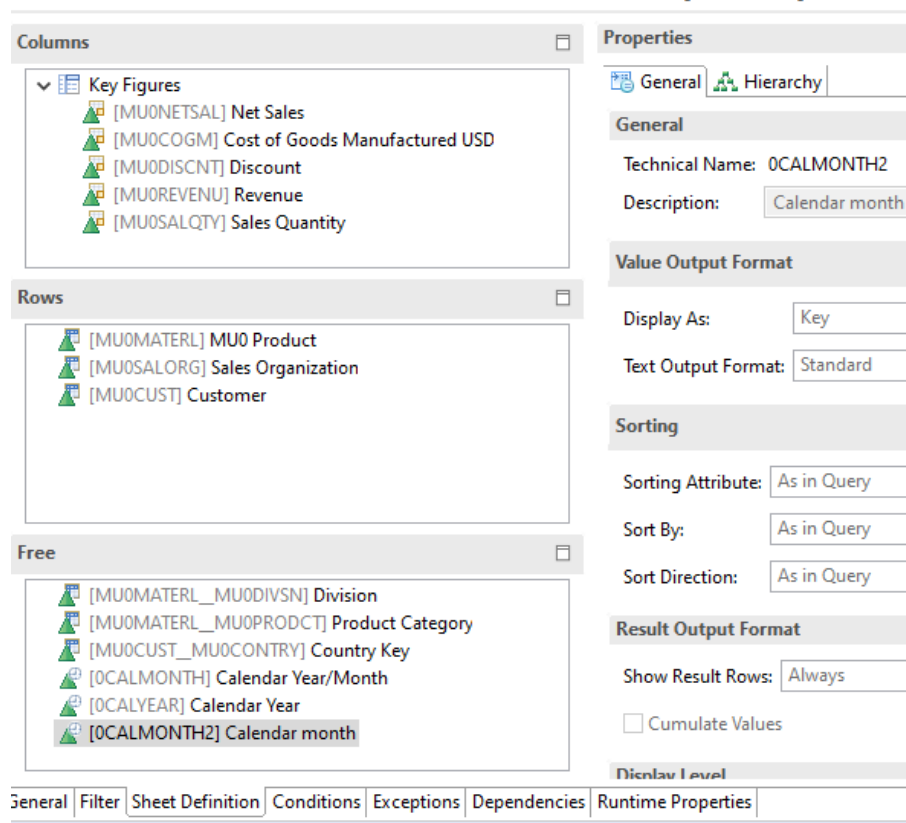


Figure 3: The Sheet Definition panel

5. Saving the query
 - a. **Save** query

- b. Click the **Reporting Preview** to see the layout of key figures and characteristics (order) as they will be displayed in a crosstab when you run the query in Analysis (later in this exercise)

[UHU] MU0CP1MQ1 - MU0 GB Sales Query for Analysis

Columns

Key Figures

Rows

MU0 Product

Sales Organization

MU0 Customer

Free Characteristics

Calendar Year/Month

Calendar Year

Calendar month

Country

Division

Product Category

Data Grid

MU0 Product	Sales Organization	MU0 Customer	Key Figures	MU0 Net Sales	MU0 Sales Quantity	MU0 Discount	MU0 Revenue	Cost of Goods Manufa
Air Pump	Germany North	Alster Cycling		103 116,08 EUR	3 928 PC	3 189,00 EUR	106 305,08 EUR	\$ 56 835,45
Air Pump	Germany North	Capital Bikes		166 921,56 EUR	6 321 PC	5 162,73 EUR	172 084,29 EUR	\$ 91 617,37
Air Pump	Germany North	Cruiser Bikes		117 068,76 EUR	4 446 PC	3 620,72 EUR	120 689,48 EUR	\$ 64 259,28
Air Pump	Germany North	Drahtesel		90 080,29 EUR	3 327 PC	0,00 EUR	90 080,29 EUR	\$ 48 136,11
Air Pump	Germany North	Fahrrpott		97 646,90 EUR	3 618 PC	0,00 EUR	97 646,90 EUR	\$ 52 332,43
Air Pump	Germany North	Ostseerad		52 996,57 EUR	1 940 PC	0,00 EUR	52 996,57 EUR	\$ 28 082,29
Air Pump	Germany North	Red Light Bikes		135 873,11 EUR	5 176 PC	4 202,31 EUR	140 075,42 EUR	\$ 74 744,02
Air Pump	Germany North	Velodrom		80 211,43 EUR	2 963 PC	0,00 EUR	80 211,43 EUR	\$ 42 871,99
Air Pump	Germany North	Result		843.914,70 EUR	31.719 PC	16.174,76 EUR	860.089,46 EUR	\$ 458.878,94
Air Pump	Germany South	Airport Bikes		113 338,78 EUR	4 344 PC	3 505,31 EUR	116 844,09 EUR	\$ 62 712,19
Air Pump	Germany South	Bavaria Bikes		223 052,44 EUR	8 657 PC	11 740,72 EUR	234 793,16 EUR	\$ 125 086,77
Air Pump	Germany South	Neckarad		118 513,46 EUR	4 506 PC	3 665,38 EUR	122 178,84 EUR	\$ 65 147,64
Air Pump	Germany South	Rikdelland		182 846,30 EUR	6 921 PC	5 655,10 EUR	188 501,40 EUR	\$ 100 296,49
Air Pump	Germany South	Result		637.750,98 EUR	24.428 PC	24.566,51 EUR	662.317,49 EUR	\$ 353.243,09
Air Pump	US East	Beantown Bikes		\$ 244 522,03	8 265 PC	\$ 12 870,00	\$ 257 392,03	\$ 119 672,32
Air Pump	US East	Big Apple Bikes		\$ 143 507,70	4 750 PC	\$ 4 438,23	\$ 147 945,93	\$ 68 755,95
Air Pump	US East	DC Bikes		\$ 122 096,39	4 038 PC	\$ 3 776,23	\$ 125 872,62	\$ 58 589,20
Air Pump	US East	Furniture City Bikes		\$ 65 181,79	2 081 PC	\$ 0,00	\$ 65 181,79	\$ 30 194,18
Air Pump	US East	Motown Bikes		\$ 124 951,11	4 119 PC	\$ 3 864,19	\$ 128 815,30	\$ 59 801,41
Air Pump	US East	Peachtree Bikes		\$ 118 458,23	3 914 PC	\$ 3 663,54	\$ 122 121,77	\$ 56 733,84
Air Pump	US East	Philly Bikes		\$ 118 759,21	3 920 PC	\$ 3 672,97	\$ 122 432,18	\$ 56 949,27
Air Pump	US East	Windy City Bikes		\$ 62 360,19	2 159 PC	\$ 1 928,56	\$ 64 288,75	\$ 30 210,66
Air Pump	US East	Result		\$ 999.836,65	33.246 PC	\$ 34.213,72	\$ 1.034.050,37	\$ 480.906,83
Air Pump	US West	Northwest Bikes		\$ 124 842,83	4 151 PC	\$ 3 861,01	\$ 128 703,84	\$ 59 950,31

Figure 4: Reporting Preview

6. Close Eclipse

Analyzing Data Using SAP Analysis

SAP Analysis for Excel can use data from SAP BW either directly from an InfoProvider (composite provider) or via a Query based on an InfoProvider. We are now ready to use the query you have designed (earlier in this exercise) to get data from our data source (composite provider) and then analyze it using SAP Analysis.

- Analyzing data using SAP Analysis, Edition for Microsoft Office, Excel
 - Go to **Start → All Programs → SAP Business Intelligence → SAP Analysis → Analysis for Microsoft Excel** (your path may be different)
 - See additional Tab entries labeled **“Analysis”** and **“Analysis Design”** (Figure 5 **Error! Reference source not found.**).

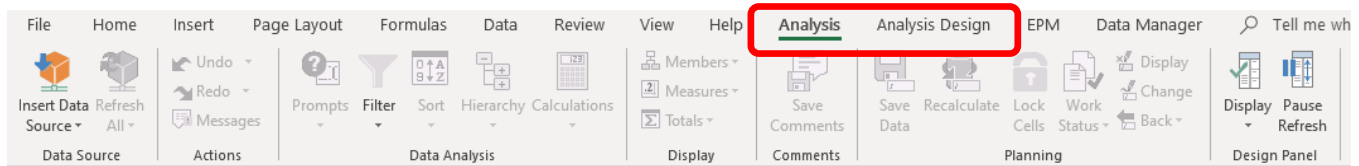


Figure 5: Analysis and Analysis Design tabs

- Click on **Analysis** tab. (If you cannot see Analysis ribbon, activate it via the **Office Button** → **Excel Options** . Go to Add-Ins and choose **Manage** → **COM Add-Ins**. Click **Go**. Make sure that “Analysis” is checked (see Figure 6 **Error! Reference source not found.**).

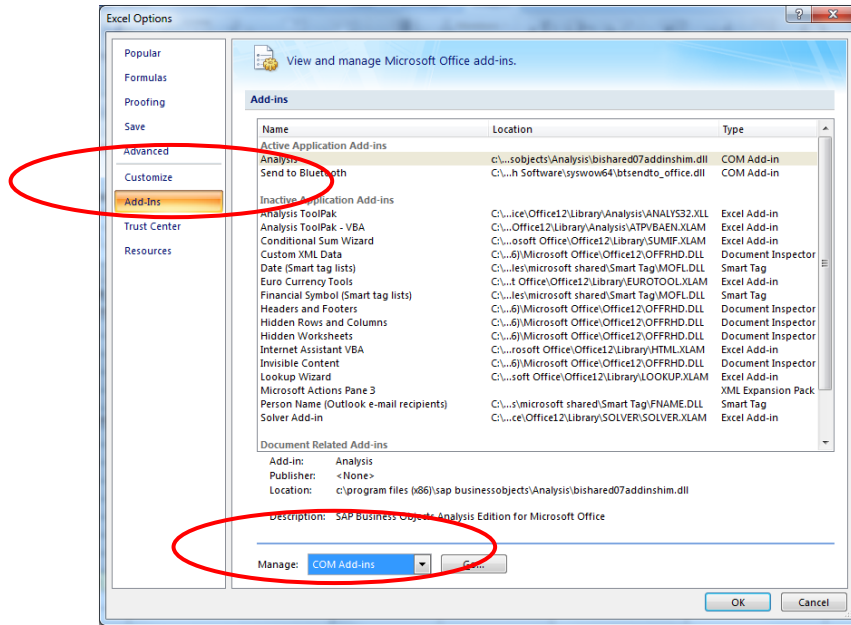


Figure 6: Enabling Analysis tab

- In the **Analysis** tab, click in Cell **A1**
- Click on **Insert Data Source** and choose **Select Data Source for Analysis...**
- If you see a window to **Logon to SAP BusinessObjects Enterprise (BOE)**, skip it. (Note: skipping BI platform means you can only store the workbooks locally on your computer or you can store them in SAP BW but not on SAP BOE).
- In the **Select Data Source** screen, choose your **SAP BW** server → **Next** → enter your login information.
- Click **OK**.
- There are two ways to search for your query. Follow *one* of the following two methods.
 - In the **Search** tab, search for **XXXCP1MQ1###**, to list your query. Select your **XXXCP1MQ1###**, and click **OK**.
- In the **Folders** tab, change the View filter to Area. Go to University Alliances BW/4HANA Curriculum → your class InfoArea → your class **XXXCP1000 Sales Data Composite Provider**. Then find your **XXXCP1MQ1###** query. Select your query and click **OK**.
- You now see the results of your query as a crosstab in SAP Analysis. Explore the various panels and display of Analysis in Figure 7.

The screenshot shows the SAP Analysis for Excel interface. The main data table is displayed with columns for Characteristics (MUO Product, Sales Organization, Customer), Key Figures/Measures (Net Sales, Cost of Goods Manufactured, Discount, Revenue, Sales Quantity), and Analysis Ribbon (Analysis Design, EPM, Data Manager, etc.). The Design Panel on the right shows the Analysis ribbon with buttons for Display, Pause, Refresh, and Design Panel. The Design Panel also shows the Analysis ribbon with buttons for Display, Pause, Refresh, and Design Panel. The Design Panel also shows the Analysis ribbon with buttons for Display, Pause, Refresh, and Design Panel.

MUO Product	Sales Organization	Customer	Net Sales	Cost of Goods Manufactured	Discount	Revenue	Sales Quantity
Air Pump	Germany North	Alster Cycling	103,116.08 EUR	56,835.45	3,189.00 EUR	106,305.08 EUR	3,928
		Capital Bikes	166,921.56 EUR	91,617.37	5,162.73 EUR	172,084.29 EUR	6,321
		Cruiser Bikes	117,068.76 EUR	64,259.28	3,620.72 EUR	120,689.48 EUR	4,446
		Drahtesel	90,080.29 EUR	48,136.11	0.00 EUR	90,080.29 EUR	3,327
		Fahrprot	97,646.90 EUR	52,332.43	0.00 EUR	97,646.90 EUR	3,618
		Ostseerad	52,996.57 EUR	28,082.29	0.00 EUR	52,996.57 EUR	1,940
		Red Light Bikes	135,873.11 EUR	74,744.02	4,202.31 EUR	140,075.42 EUR	5,176
		Velodrom	80,211.43 EUR	42,871.99	0.00 EUR	80,211.43 EUR	2,963
		Result	843,914.70 EUR	458,878.94	16,174.76 EUR	860,089.46 EUR	31,719
	Germany South	Airport Bikes	113,338.78 EUR	62,712.19	3,505.31 EUR	116,844.09 EUR	4,344
		Bavaria Bikes	223,052.44 EUR	125,086.77	11,740.72 EUR	234,793.16 EUR	8,657
		Neckarrad	118,513.46 EUR	65,147.64	3,665.38 EUR	122,178.84 EUR	4,506
		Riedelland	182,846.30 EUR	100,296.49	5,655.10 EUR	188,501.40 EUR	6,921
		Result	637,750.98 EUR	353,243.09	24,566.51 EUR	662,317.49 EUR	24,428
	US East	Beantown Bikes	\$ 244,522.03	\$ 119,672.32	\$ 12,870.00	\$ 257,392.03	8,266
		Big Apple Bikes	\$ 143,507.70	\$ 68,755.95	\$ 4,438.23	\$ 147,945.93	5,038
		DC Bikes	\$ 122,096.39	\$ 58,589.20	\$ 3,776.23	\$ 125,872.62	4,398
		Furniture City Bikes	\$ 65,181.79	\$ 30,194.18	\$ 0.00	\$ 65,181.79	2,081
		Motown Bikes	\$ 124,951.11	\$ 59,801.41	\$ 3,864.19	\$ 128,816.70	4,119
		Peachtree Bikes	\$ 118,458.23	\$ 56,733.84	\$ 3,663.54	\$ 122,121.61	3,914
		Philly Bikes	\$ 118,759.21	\$ 56,949.27	\$ 3,672.97	\$ 122,431.18	3,920
		Windy City Bikes	\$ 62,360.19	\$ 30,210.66	\$ 1,928.56	\$ 64,288.75	2,159
		Result	\$ 999,836.65	480,906.83	\$ 34,213.77	\$ 1,034,050.37	33,246
	US West	Northwest Bikes	\$ 124,842.83	\$ 59,950.31	\$ 3,864.19	\$ 128,703.84	4,151
		Rocky Mountain Bikes	\$ 126,360.52	\$ 60,528.71	\$ 3,864.19	\$ 130,266.55	4,171
		Silicon Valley Bikes	\$ 145,445.65	\$ 71,996.35	\$ 3,864.19	\$ 153,100.58	5,155
		SoCal Bikes	\$ 122,239.28	\$ 58,579.63	\$ 3,864.19	\$ 126,019.81	4,377
		Windy City Bikes	\$ 66,476.51	\$ 31,577.00	\$ 1,928.56	\$ 68,532.49	2,159
		Result	\$ 585,364.79	282,777.00	\$ 21,260.48	\$ 606,625.27	20,505
		Result	\$ 1,439,276.47	\$ 741,814.82	\$ 65,444.24	\$ 1,440,714.74	46,998
City Bike Max	Germany North	Alster Cycling	257,257.03 EUR	128,628.51	7,956.44 EUR	265,213.47 EUR	1,176
		Capital Bikes	324,488.70 EUR	162,244.35	10,035.74 EUR	334,524.44 EUR	1,226
		Cruiser Bikes	259,691.12 EUR	129,845.56	8,031.70 EUR	267,722.82 EUR	1,189
		Drahtesel	213,586.00 EUR	106,793.02	0.00 EUR	213,586.00 EUR	1,146
		Fahrprot	308,800.00 EUR	154,400.00	0.00 EUR	308,800.00 EUR	1,206
		Ostseerad	51,181.50 EUR	25,590.75	0.00 EUR	51,181.50 EUR	1,049
		Red Light Bikes	171,403.07 EUR	85,701.53	7,061.71 EUR	178,464.78 EUR	1,162
		Velodrom	114,193.04 EUR	57,096.52	0.00 EUR	114,193.04 EUR	1,109
		Result	1,335,076.10 EUR	662,514.63	33,085.59 EUR	1,368,191.28 EUR	1,263
	Germany South	Airport Bikes	210,410.73 EUR	105,205.36	8,957.20 EUR	219,367.93 EUR	1,198
		Bavaria Bikes	377,688.95 EUR	188,844.47	25,700.20 EUR	403,433.15 EUR	1,360

Figure 7: Analysis for Excel layout

11. Setting your own default **display style** for Analysis for the graphic look of your results.

- You can apply a new **Style Set** and set it as default.
- In the Analysis Design ribbon, under **Styles** → **Apply Style Set**, choose **SAP Tradeshow Plus** and **Set as Default**. Click **OK** (you may choose another style of your choice).
- You can also design custom styles sets

12. Exploring the Design Panel

- Open the **Design Panel** if not already open using the **Design Panel** → **Display** button in the **Analysis** ribbon.

- b. **Analysis** tab → You can display and change the navigation status of your query results. You can drag and drop Measures and characteristics into Columns and rows as desired.
 - c. **Information** tab → You can display information about the embedded data source or the workbook itself
 - d. **Components** tab → You can display the structural view either from the data source or the worksheet perspective
 - e. **Design Rules** tab → You can display and maintain the rules created with Table Design. The rules are displayed for each data source and rule type.
 - f. **Comments** tab → You can see a section that shows the comment history for a selected cell and a section that shows the context information for the cell.
13. Keep Analysis open for the next task.

Slicing and Dicing within Analysis

In order to derive insights from the vast amount of data in the data source, it is essential to understand **Online Analytical Processing (OLAP)** analysis. This includes navigation within the data using **Design Panel** and the query results area (also called Cross tab).

1. Task: Currency Translation

1. GB is a global company and deals with multiple currencies. During the modeling of the GB aDSO, we chose the key figure CoGM with a **fixed currency** of USD. Therefore, in the crosstab you will notice that Cost of Goods Manufactured is displayed in USD. The other measures – discount, net sales, and revenue are in different currencies in different sales organizations. Notice that the total discount for all sales organizations is displayed as ‘*’. This is because of mixed currencies for that column. You will now do a currency translation
 - a. Click on **Measures** → **Currency Translation** in the Analysis ribbon
 - b. Conversion routines have already been configured in SAP BW/4 HANA. One can convert to either **USD** or to **EURO** target currency. Choose **Target Currency: USD, Currency Conversion Type: BW/4HANA 2.0 Curriculum: Fixed USD** (this routine has been preconfigured for you. Many other routines can be created for doing currency translation)
 - c. Click **OK**.
 - d. **CoGM, Discount, Net Sales** and **Revenue** are all now in USD.
 - e. The totals are now displayed in USD instead of ‘*’

Question 1: What is the overall Revenue in USD?

2. Task: Basic Navigation

1. The **Analysis** tab in the **Design** Panel is used to add, remove, order, and filter the cross tab as desired. You notice that the query based on a source is now displayed in simple business terms. Prior *pivot table* skills will come in handy for the rest of these exercises.
2. Switch to the **Analysis** tab in the **Design** Panel
3. Ordering Measures
 - a. In the **Columns** subpanel, change the order of the **Measures** to – **Sales Quantity**, **Revenue**, **Discount**, **Net Sales** and **CoGM**. *Note the impact of changing the order of your measures in the cross tab area.*
4. Ordering Characteristics
 - a. Add **Calendar Year** to the drilldown by using drag & drop from available field to the **Rows** area.
 - b. Remove **Sales Organization** and **Customer** from the Rows area by right clicking.
 - c. Change the order of **Rows** to – **Calendar Year**, **Product**. See the impact of changing the order in your crosstab

Question 2: What is the Revenue for E-Bike in 2010?**Question 3: What is the Revenue for E-Bike in 2009?**

5. Add Customer back into Rows.

3. Task: Saving Your Workbook

1. There are several options for saving your workbook. You can save the workbook locally on your computer or on the SAP BW/4HANA server
 - a. For this exercise, **save** your workbook locally after each task.
 - b. You can reopen the workbook on your computer at a later time and continue from there.

4. Task: Sorting

1. We can sort, rank, and filter the crosstab query in various ways.

2. Sorting by Characteristics

- a. Choose **Calendar Year** in the cross tab. Then **Sort → Ascending** (or you can right click. It may already be sorted in ascending). You see **2007** listed before **2008, etc.**
- b. Characteristics often have both a **key** column and a **text** column. Sorting applies to whichever column you choose.
- c. We would like to display both the Product **Key** and **Text**. Right click **Product → Members → Key and Text**. You now see the **Product key** (e.g., PUMP100) and **text** (e.g., Air Pump).
- d. Now sort **Product Key** in **Descending**. You should see Products list in descending order of their numbers (SHRT1000 before RKIT1000). Sorting of Calendar Year should be unaffected.

Note: To know the sorting status of any characteristics, right click the column name. The current sorting status is greyed out. You may choose the other available sorting order.

Question 4: What is the Sales Quantity of T-shirts for Airport Bikes in 2007?

3. Sorting by Measures

- a. Add **Customer** to rows.
- b. Choose the Measure **Sales Quantity**. **Sort → Descending**
- c. You see that the Sales Quantity has been sorted by the sales quantity *subtotals* in this order: Calendar Year, then Product and then Customer. i.e. the total Sales Quantity for each year is sorted in descending order, within that year by product, and then within that product by customer.
- d. To illustrate this better, **remove** the characteristics **Product** and **Customer**

Question 5: What year had the highest Sales Quantity?

Question 6: What year had the lowest Revenue?

4. Now add **Product** (after Calendar Year) to rows

Question 7: In the year with the lowest Revenue, which Product had the lowest revenue?

Question 8: Which Product had the lowest revenue overall?

5. Complex sorting by Characteristics

- a. **Add** the Characteristics in this order – **Calendar Year**, **Division**, **Customer** (remove the other characteristics)
- b. Choose the **Calendar Year** Characteristic (in the crosstab). **Sort** → **More Sort Options** (in the ribbon for Analysis).
- c. Here you can Sort by **Default**, **Member Display Type**, **Attribute** or **Measure**.
 - i. **Default (Dimension Members by First Display Type)**: The members are sorted by the first display type (key or text) of the selected dimension (Characteristic).
 - ii. **Member Display Type**: You can select one of the member display types. E.g. key, text etc.
 - iii. **Attribute**: If the characteristic has attributes, you can sort by any one of those attributes. E.g. *Color* attribute of the Product Characteristic. This option will not be listed if the characteristic does not have attributes
 - iv. **Measure**: Sort is applied to the values of the selected measure (e.g. Sales Quantity).
.e.g. Sort customer by descending revenue.
- d. Choose **Measure** → **Net Sales**, **Sort Descending**. **OK**.
- e. Then, choose **Division** and **sort it in descending order of Net Sales**.
- f. Then, choose **Customer** and **sort it in descending order of Net Sales**.

Question 9: In the year that had the highest Net Sales, what Division had the highest Net Sales?

Question 10: For the division in Question 9, which customer had the highest Net Sales (remove year)?

Question 11: In the year with the lowest Revenue, what division had the highest revenue?

6. Remove **Year** from Rows

Question 12: For the Division in Question 11, which Customer had the highest Net Sales overall?

5. Task: Filtering

1. We can filter the crosstab by Characteristic values and/or by measure values to limit the data displayed.
2. Rows: Calendar Year, Division, Customer.
3. Filtering by Characteristic

- a. We want to filter the Customer to the values **Bavaria Bikes** and **Beantown Bikes**
 - i. In the **Data Source** fields (left side of Display Panel) expand **Customer** to display the Characteristic members.
 - ii. Select **Bavaria Bikes** and **Beantown Bikes** together using the **CTRL** key and drag both elements into the **Rows** area under Customer.
 - b. The **filter** icon next to the **Customer** Characteristic indicates that you have selected filter values. Only two customers are included in your query results.
 - c. Move **Customer** from the **Rows** to the **Background Filter** area. What happens to the results table? **Customer** is removed from the table but the filter elements are still valid.
 - d. Move **Customer** back to **Rows** (after Division).
4. Pausing (automatic) refreshing
 - a. The **Pause Refresh** button gives you the option of pausing the refresh (roundtrip to datasource) while you navigate. Doing so reduces the refresh time until you are ready to display results.
 - b. Click **Pause Refresh**.
 5. Use the **Filter By Member** option from the **context menu** for **Calendar Year** (in Rows) to show **2009** only. Notice that the query results are not refreshed.
 6. Working with Characteristics Attributes
 - a. Display the attribute **City** for Customer.
 - b. In the **Data Source Fields** expand **Customer** and expand the Attributes node
 - c. Select **City** and drag it on to Customers in the rows area
 - d. Click **Pause Refresh** thereby disabling it. Your query results are refreshed.
 - e. Notice how attributes (**City**) are handled in the query results.
 - f. Remove the filter on the customer Characteristic

Question 13: What Customer provided the highest Revenue for accessories (Division is AS) in 2009?

Question 14: What city provided the highest Sales Quantity for accessories in 2009 (look carefully)?

7. Ranking and Filter by Measure
 - a. You can define rules to filter the data. You can apply one or multiple rules to a measure. Depending on where you open the **Filter by measure** dialog, the filter definition is applied to a selected Characteristic, to all Characteristics or to the most detailed

- Characteristic in your analysis. The applied measure filters do not affect *totals* or *subtotals* in your analysis.
- b. Remove the filter on **Calendar Year** so that all years are listed. Remove the filter on **Division**. Also remove the attribute **City**.
 - c. Choose **Revenue**. **Filter** → **Filter by Measure**.
 - d. You see three options:
 - i. **All Dimensions Independently**: The filter is applied to all dimensions in the analysis, from the outermost to the most detailed one. For example, in an analysis with three dimensions in columns A, B and C, the filter is first applied to the dimension in column A, then to the dimension in column B and finally to the dimension in column C
 - ii. **Most Detailed Dimension in Rows**: The filter is applied to the most detailed dimension of the rows. For example, in an analysis with three dimensions in columns A, B and C, the filter is applied to the dimension in column C.
 - iii. **Most Detailed Dimension in Columns**: The filter is applied to the most detailed dimension of the columns. For example, in an analysis with three dimensions in rows 2, 3 and 4, the filter is applied to the dimension in row 4.
 - e. Choose **All Dimensions Independently** → **Edit**.
 - f. **Based on Measure** = **Revenue**. **Definition** = **Top N, 3** → **Add** → **OK**.
 - g. You see the top 3 Years, followed by top 3 Divisions and top 3 Customers, all by Revenue.

Question 15: What is the top Year in terms of Revenue (for the Top 3)? What is the top Division in the top Year in terms of Revenue?

8. Choose **Revenue**. **Filter** → **Filter by Measure** → **All Dimensions Independently** → **Reset**. The top 3 filter is removed.
9. Choose **Revenue**. **Filter** → **Filter by Measure** → **Most Detailed Dimension in Rows** → **Edit**.
 - a. Define a rule that lists the top 3 Revenue
 - b. You should now see the top three Customers for each Division

Question 16: In 2011, for bicycles (Division is BI), what is the total Revenue for the top three Customers?

10. Keep the workbook open for the next task.

Task: Totals and Sub-totals

1. Remove the top 3 filter applied above (on **Revenue**)
2. We would like to change the aggregation for measures from the default **SUM** (set in SAP BW/4HANA) to one of our choice.
 - a. Choose **Revenue**. Then **Totals** → **Calculate Total As** → **Average**.

Question 17: What is the average Revenue per Customer per Year per Division?

Question 18: What is the average Revenue per Customer?

3. Save. Keep the workbook open for the next task.

Task: Advanced Sorting and Filtering

Efficient sorting and filtering is essential for getting quickly to data that we want to see.

1. We would like to list only the *top 2 Customers* (in terms of **Revenue**) for each **Calendar Year** and **Product** combination.
 - a. Start with these characteristics in rows – **Calendar Year**, **Product**, **Customer**
 - b. Change the aggregation of **Revenue** totals to **default** by right-clicking on **Revenue**.
 - c. Right-click **Revenue** and then **Filter by Measure** → **Most Detailed Dimension in Rows** → **Edit**.
 - d. Choose **Revenue** for **Based on Measure**. In **Definition**, choose **top N**, **2** → **Add** → **OK**.
 - e. You should see the top 2 **Customers** for each **Product** within each **Calendar Year**.
2. We would like to list the *top 2 Customers* (irrespective of the product and year) and display the **Product** and **Year** they occurred in.
 - a. Remove the filter by right-clicking **Revenue** → **Filter by Measure** → **Most Detailed Dimension in Rows** → **Reset**.
 - b. Click **Revenue**. Then **Sort** → **More Sort Options**. **Sort Descending**. Select **Break Hierarchies** → **OK**.

Break Hierarchies: *In the default setting of the crosstab, data is sorted in the hierarchy levels of the crosstab. Parent members are sorted in order, and child members are sorted below each parent member in their own order. If you want to sort data across the entire hierarchy, select the "Break Hierarchies" option.*

- c. Now you see that the **Customers** are listed in descending order of **Revenue** with their respective **Years** and **Products**.

Question 19: For which Year and Product did the highest Revenue from a single Customer occur?

- d. Right-click **Revenue**, and then **Filter by Measure** → **Most Detailed Dimension in Rows** → **Edit**.
 - e. Choose **Revenue** for **Based on Measure**. In **Definition**, choose **top N**. Enter **2** → **Add** → **OK**.
 3. **Save**. Keep the workbook open for the next task.

Task: Conditional Formatting

Conditional formatting is used to highlight important values or unexpected results in your data. With conditional formatting, you define *rules* to select different or critical values in your query results. Results that fall outside a set of predetermined threshold values are highlighted in color or designated with symbols. This enables you to identify immediately any results that deviate from the expected results. You can add one or more rules to a conditional format and prioritize the application of the rules.

1. We will reset some of the sorting and filtering from the previous task
 - a. Start by **removing the top 2 filter** on **Revenue**
 - b. Sort **Calendar Year** in ascending order
 - c. Sort **Product** in ascending order
 - d. Sort **Customer** in ascending order
2. Conditional Formatting
 - a. Go to **Conditional Formatting** (Analysis Design tab) to create **New Conditional Formatting**.
 - b. Create the **Rules** as shown in Figure 8. You will need to create three rules. Enter the revenue bounds for the rules as shown.

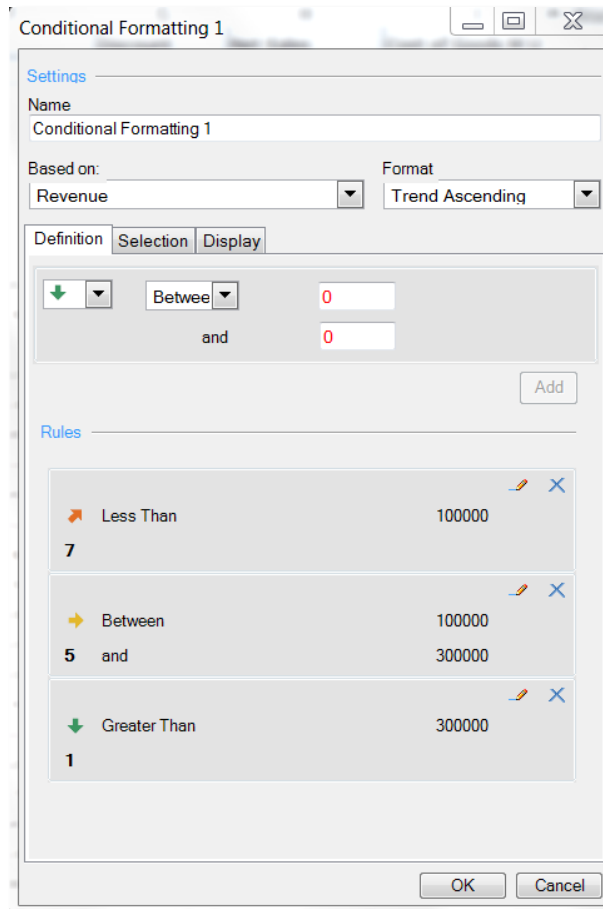


Figure 8: Conditional Formatting

- c. The **Revenue** measure is conditionally formatted using the rules you have created.
 - d. This gives you visual indication of your underperforming, adequate, or valued customers and products etc. based on measures such as Revenue. Look for green arrows as indicators that the measured value is acceptable and red arrows as values that indicate we should look more closely at the associated data.
3. **Save.** Keep the workbook open for the next task.

9. Task: Charting

You will now create a chart based on the query results

1. First, delete conditional formatting from the previous task.
2. Remove **Customer** and **Product** from **Rows**.
3. Right-click all **Result** rows and **hide Totals**. Do the same for Overall Result. This can also be done by right-clicking on the associated columns.
4. Click anywhere within the crosstab.

5. Click on **Chart** to insert a chart based on your query results
 - a. Move the Chart below your crosstab.
 - b. You can double-click the graph to choose graph properties.
 - c. Change the Chart Type to any of your choice
 - d. Select a layout of your choice. Select a Chart Style of your choice.
 - e. Remove all Measures except **Revenue**.
 - f. Put only **Calendar Year** and **Country** in the Rows.
 - g. Change the Chart Type to a **Line Chart**
 - h. Move the **Country** characteristic to Columns (after Measures)
 - i. You should now see the revenue for DE by year, US by year and Overall by Year (in the crosstab).
 - j. You can see the revenue trend in the two countries from 2007-2019

10. Task: Creating a PowerPoint Slide

You have inserted a data source in Analysis, edition for Microsoft Excel. Now, you can create a slide with the query that is currently displayed in the workbook. In PowerPoint, the current navigation state of the objects in Excel is displayed, but the objects exist independently in Excel and PowerPoint.

1. To generate a PowerPoint slide, you can click on **Smart Copy** in the Analysis Design ribbon of Excel (Figure 9).

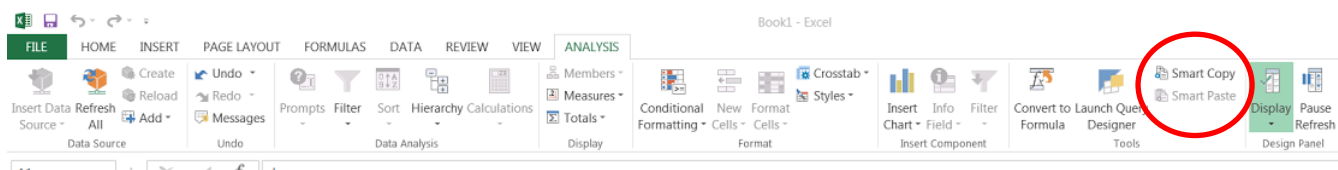


Figure 9: Creating PowerPoint slide

- a. You can now copy your analysis into PowerPoint or other document types.
 - b. Alternatively, open **Analysis for Microsoft PowerPoint**.
 - c. Open a new blank presentation.
 - d. Click on the Analysis tab in PowerPoint.
 - e. Then click **Smart Paste**.
 - f. You can now use sorting, filtering, and other OLAP techniques described earlier in this homework.
2. **Save** the PowerPoint presentation. Close PowerPoint.

11. Task: GB Trend Analysis

1. Using the techniques explored in Tasks 1 through 9, answer the following questions. These are open ended questions whose objective is to explore, exploit and discover some of the trends in the GB sales data.
2. Use **charts** to visualize trends, spot outliers, discover sudden changes.

Question 20: What Year had the highest Revenue? What was the Revenue?

Question 21: What product had the highest Revenue in total? What was the Revenue?

Question 22: Are the historical (year by year) Revenue trends for the US and DE similar or dissimilar?

Question 23: Did GB ever gain or lose a Customer (based on Revenue)? Explain.

Question 24: Is there seasonality in Revenue during the year? If so, what Month has the highest Revenue? Is the seasonality similar from year to year?

Question 25: For what Year and Product did the highest Revenue from a single Customer occur?

Question 26: Are there any products that show dramatic change in Revenue over time (years)? Does it have the same change by Country?

Question 27: Is there any Product(s) that does not display significant seasonality?

Question 28: What customer has the highest percentage contribution to Revenue? What has been the trend of that Customer's percentage contribution over the years? Hint: Use Calculations.